

OFFLINE DEMO BUILD · NO REAL ORDERS

Bank Nifty Futures Trading App

Step-by-Step Guide — how it works, locally, from start to finish

આ માર્ગદર્શિકા તમને સમજાવે છે કે આ એપ શું કરે છે, અંદરની ફાઇલોનું કામ શું છે, અને દરેક સ્ટેપ કેવી રીતે ચલાવવો — સ્ક્રીનશોટ સાથે.

What is inside: a full walk-through of `setup` → `validate` → `check` → `demo` → `backtest` → `report` → `web` → `run` → `preflight` → `go-live`, with real screenshots captured from a local offline run.

Not financial advice. This is an execution framework. It can place real orders in live mode. Test in *paper* mode and on a backtest first.

Generated from a local offline demo (synthetic data, paper execution). Date: 2026-10-04

Contents

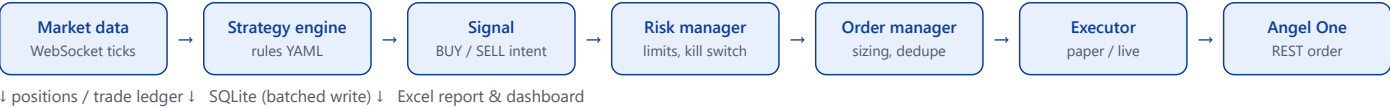
1. What this app does & the pipeline
2. Project folder map — which file does what
3. Setup — install, `.env`, credentials
4. Step 1 — `validate` (check the strategy)
5. Step 2 — `check` (broker connectivity, read-only)
6. Step 3 — `demo` (offline run through the live pipeline)
7. Step 4 — `backtest --synthetic` (strategy-only replay)
8. Step 5 — `report` (Excel of every trade)
9. Step 6 — `web` console tour (all 6 tabs)
10. Step 7 — `run` paper trading (what happens every tick)
11. Step 8 — Go-live safety (preflight, arm, kill switch)
12. Step 9 — End-to-end trade walkthrough + the 15 rules
13. Troubleshooting & FAQ

1. What this app does & the pipeline

An event-driven Python app that trades **Bank Nifty near-month futures** against **Bank Nifty Spot** signals. The same engine runs live trading and backtesting, so the two can never drift apart. Strategies are plain **YAML files** — you change rules without rewriting code.

આ એપ **Bank Nifty futures** ને Angel One SmartAPI દ્વારા ઓટોમેટિક ટ્રેડ કરે છે. શરૂઆતમાં તે **paper** મોડમાં ચાલે છે — સાચા ભાવ મળે છે, પણ ખરેખર ઓર્ડર મોકલાતો નથી. બધા નિયમો **YAML ફાઇલ**માં લખેલા છે, એટલે કોડ બદલ્યા વગર સ્ટ્રેટેજી બદલી શકાય.

The data pipeline (every tick travels this path)



Hot path = fast

- The engine does **in-memory math only** on the hot path: no DB reads, no Excel, no blocking I/O.
- Ticks arrive on a WebSocket thread and are handed to the asyncio loop.
- SQLite writes happen on a separate **batched writer thread**.

Two modes

- **Paper (default):** real Angel One prices, simulated fills, zero real orders. Only the final broker call is swapped out.
- **Live:** real orders, real money — behind a three-flag interlock.

Key idea. Paper mode is not a toy. The strategy, risk gates, position book and P&L you watch are the real ones — only the fill is simulated.

મુખ્ય વાત: Paper મોડ નકલી નથી. સ્ટ્રેટેજી, રિસ્ક, પોઝિશન અને P&L બધું સાચું ચાલે છે — ફક્ત ઓર્ડર સાચો મોકલાતો નથી.

2. Project folder map — which file does what

નીચે પ્રોજેક્ટના ફોલ્ડરનું કામ ટૂંકમાં લખ્યું છે. દરેક ફોલ્ડરનું એક જ જવાબદારી છે.

```
Angelone app/
├─ main.py           convenience shim -> python main.py <command>
├─ README.md         full project docs
├─ LIVE_TRADING.md   paper -> live operating manual
├─ .env              your secrets + settings (git-ignored)
├─ strategy_versions/ v1_baseline.yaml, v2_example.yaml (the RULES)
├─ reports/          generated Excel reports land here
├─ data/             SQLite DB, users, instruments, kill-switch file
├─ logs/             app log + dedicated trade log
├─ banknifty_trading_app/
│   └─ the application package
│       ├── main.py   CLI entry point (all commands)
│       ├── runtime.py LIVE wiring: feed -> engine -> risk -> executor
│       ├── config.py typed settings + the live interlock
│       ├── preflight.py go-live checklist (read-only)
│       ├── selftest.py `check` connectivity self-test
│       ├── demo.py     offline synthetic run through the LIVE pipeline
│       ├── credentials.py reads/writes .env safely
│       ├── core/       events, bus, models, in-memory state
│       ├── angelone/   auth, REST, WebSocket, instruments (SmartAPI)
│       ├── market_data/ feed, tick store, daily closes, staleness
│       ├── strategy/   engine + indicators (the brain)
│       ├── rules/      base, registry, filters, entries, stops, targets
│       ├── risk/       manager, limits, kill switch
│       ├── execution/  order manager, router, paper, live, recovery
│       ├── positions/  book, ledger, P&L
│       ├── backtest/   replay, run, data loader, costs, synthetic data
│       ├── database/   SQL DDL, session, batched writer, repository
│       ├── reports/    Excel writer
│       ├── dashboard/  minimal read-only dashboard
│       ├── web/        full read-write web console (FastAPI + JS)
│       ├── logging/    logging setup
│       └─ tests/       pytest suite
```

Rule of thumb: to change *what* the strategy does, edit YAML in `strategy_versions/`. To change the *engine*, edit `strategy/engine.py` or add a rule in `rules/`. Everything else (broker, data, DB) stays untouched.

3. Setup — install, `.env`, credentials

પહેલા Python virtual environment બનાવો, પછી ડિપેન્ડન્સી ઇન્સ્ટોલ કરો, અને `.env` ફાઇલમાં તમારી Angel One માહિતી ભરો.

3.1 Create the environment & install dependencies

```
PowerShell

python -m venv .venv
.\.venv\Scripts\python.exe -m pip install -r requirements.txt

copy .env.example .env      # then fill in your Angel One credentials
```

The offline commands (`demo` , `backtest --synthetic` , `report` , and the `web` console) need **no broker credentials**. Only `check` , `paper run` and `live` need them.

3.2 Fill in `.env`

```
.env (never commit this file)

MODE=paper
LIVE_TRADING=false
LIVE_ARMED=false          # master interlock - leave false until you go live
ANGEL_API_KEY=...
ANGEL_CLIENT_CODE=...
ANGEL_PIN=...
ANGEL_TOTP_SECRET=...
```

You can also let the app write the secrets for you (input is not echoed):

```
PowerShell

python -m banknifty_trading_app.main setup
```

અથવા `setup` કમાન્ડ વાપરો — તે તમારાથી માહિતી વાંચે સીધું `.env` માં લખે છે (ટાઇપ કરેલું સ્ક્રીન પર દેખાતું નથી).

Where to get the values: SmartAPI portal → Create App (API key), your Angel One login id (client code), 4-digit trading PIN, and the **TOTP secret seed** (the base32 seed, not the rotating 6 digits).

4. Step 1 — validate

Shows the **resolved** strategy and config: what the engine will actually run, before you start anything.

આ કમાન્ડ બતાવે છે કે એપ કઈ સ્ટ્રેટેજી અને કઈ સેટિંગ વાપરવાની છે — ટ્રેડિંગ શરૂ કરતાં પહેલાં ખાતરી કરવા માટે.

```
PowerShell: python -m banknifty_trading_app.main validate

Strategy: v1_baseline
description : Original Bank Nifty strategy: 20 SMA filter on daily Spot closes; entry on a fre
instrument  : BANKNIFTY near_month x2 lots (30/lot)
reference   : offset 2 day(s) before yesterday
SMA period  : 20
filters     : ['sma']
entry       : reference_cross
stop        : daily_close
partial     : percent_move
trailing    : percent_move_placeholder
expiry      : always
reversal    : on x2

Mode: paper (live trading disabled)
Credentials: {'angel_api_key': 'missing', 'angel_client_code': 'missing', 'angel_pin': 'missing', 'angel_totp_secret': 'missing'}
```

How to read it: `instrument` = what it trades (BANKNIFTY futures, 2 lots of 30). `reference` = the cross level is the close from 2 days before yesterday. `filters` = the 20 SMA filter. `entry` / `stop` / `partial` / `trailing` / `expiry` / `reversal` = the rule actually loaded for each stage. `Mode: paper` means live is disabled.

5. Step 2 — `check` (read-only)

A connectivity self-test: login, LTP, daily candles and the WebSocket. It **never places an order**.

```
PowerShell: python -m banknifty_trading_app.main check

python -m banknifty_trading_app.main check
# -> logs in to Angel One, fetches an LTP + daily closes, then listens
#     for WebSocket ticks for a few seconds.
# -> NO orders are placed.
```

Needs credentials. Run this on the machine that will trade, during market hours. If the WebSocket step fails, check whether the market is open — it usually still passes with a stale tick. Corporate TLS-inspecting proxies are handled automatically (see Troubleshooting).

આ કમાન્ડ ફક્ત ચેક કરે છે કે Angel One સાથે જોડાણ થાય છે કે નહીં. કોઈ ઓર્ડર મોકલાતો નથી. તેના માટે credentials જરૂરી છે.

6. Step 3 — demo (offline, no broker)

Feeds **synthetic** market data through the **real** live pipeline: strategy engine, risk manager, order manager, paper executor, position book, trade ledger, SQLite writer, Excel report. No credentials, no broker.

આ કમાન્ડ બ્રોકર વગર, બનાવેલા (synthetic) ડેટા પર પૂરી પાઇપલાઇન ચલાવે છે. એટલે તમે credentials વગર જ આખી સિસ્ટમ ચાલતી જોઈ શકો.

```
PowerShell: python -m banknifty_trading_app.main demo --start 2025-09-01 --end 2026-03-01

2026-10-04 09:58:10 | INFO | banknifty.tls | TLS verification uses the OS trust store (truststore).
2026-10-04 09:58:10 | INFO | banknifty.demo | DEMO (offline, paper execution) strategy=v1_baseline
2026-10-04 09:58:11 | INFO | banknifty.strategy.engine | [day_start] 2025-09-29 ref=51070.96 prev=51211.21
2026-10-04 09:58:11 | INFO | banknifty.strategy.engine | [enter] BUY 2 lot(s) reason=entry_signal spot=51088.14
2026-10-04 09:58:11 | INFO | banknifty.strategy.engine | [stop_loss] BUY stop hit at spot=51052.95
2026-10-04 09:58:11 | INFO | banknifty.strategy.engine | [enter] SELL 2 lot(s) reason=reversal spot=51052.95
2026-10-04 09:58:11 | INFO | banknifty.strategy.engine | [stop_loss] SELL stop hit at spot=51223.05
2026-10-04 09:58:11 | INFO | banknifty.strategy.engine | [enter] BUY 2 lot(s) reason=reversal spot=51223.05
2026-10-04 09:58:11 | INFO | banknifty.strategy.engine | [stop_loss] BUY stop hit at spot=51196.17
2026-10-04 09:58:11 | INFO | banknifty.strategy.engine | [enter] SELL 2 lot(s) reason=reversal spot=51196.17
[ ... the engine repeats this every tick, every trading day ... ]
2026-10-04 09:58:13 | INFO | banknifty.strategy.engine | [stop_loss] SELL stop hit at spot=46165.11
2026-10-04 09:58:13 | INFO | banknifty.strategy.engine | [enter] BUY 2 lot(s) reason=reversal spot=46165.11
2026-10-04 09:58:13 | INFO | banknifty.strategy.engine | [stop_loss] BUY stop hit at spot=46160.99
2026-10-04 09:58:13 | INFO | banknifty.strategy.engine | [enter] SELL 2 lot(s) reason=reversal spot=46160.99
2026-10-04 09:58:13 | INFO | banknifty.reports.excel | Wrote report: pdf_assets\capture\reports\demo_report.xlsx
2026-10-04 09:58:13 | INFO | banknifty.demo | DEMO complete: 46 legs, realized P&L -31335.00,
```

What happened: over the 6-month window the engine generated **46 trade legs**, wrote every order / fill / trade / event to the SQLite DB, and produced `demo_report.xlsx`. On this sample the realized P&L was **-31,335** — a loss, which is exactly why you run a backtest and paper trade *before* risking money.

અહીં 6 મહિનામાં 46 ટ્રેડ થયા અને એક Excel રિપોર્ટ બન્યો. આ સેમ્પલમાં ખોટ (loss) આવી — તેથી જ પહેલા backtest અને paper trading કરવું જરૂરી છે.

7. Step 4 — `backtest --synthetic`

A **one-year** replay of the strategy on generated data, and an Excel report of every trade, equity curve and event.

```
PowerShell: python -m banknifty_trading_app.main backtest --synthetic --start 2025-09-01 --end 2026-03-01
```

```
Backtest complete: 46 legs, net P&L -45,297.15, report in pdf_assets\capture\reports
```

Demo vs backtest — important difference.

demo runs the strategy *and* the live execution path (paper broker, risk, position book, DB).

backtest exercises the strategy engine only, with a fill model and charges. Use it to judge the rules; use **demo** to watch the plumbing.

Historical (non-synthetic) mode uses real Angel One candles — Spot for signals and a stitched near-month futures series for fills — and caches chunks under `data/history/`. That mode needs credentials and is subject to the historical API's look-back limits.

Backtest = ફક્ત સ્ટ્રેટેજીની ટેસ્ટ (એક વર્ષ). Demo = આખી લાઇવ પાઇપલાઇનની ટેસ્ટ. બંને એક જ નિયમો વાપરે છે.

8. Step 5 — report (Excel from the database)

Regenerates an Excel workbook from whatever is stored in SQLite — every trade, the summary metrics, the equity curve and the strategy events.

```
PowerShell: python -m banknifty_trading_app.main report

Wrote report: pdf_assets\capture\reports\live_trades_report.xlsx
```

Report — Summary sheet (from a real backtest run)

Bank Nifty Backtest - v1_baseline	None
strategy	v1_baseline
intrabar	conservative
fills	115
charges	33316.87
net_pnl	84993.08
lot_size	30
data	synthetic
start	2025-10-04
end	2026-10-04
strategy_version	v1_baseline
description	Original Bank Nifty strategy: 20 SMA filter on daily Spot closes; entry on a fresh cross of the day-before-yesterday Spot close; 2 lots; yesterday's Spot close as the ratcheting stop; 1.7% partial on one lot; 2-lot reversal on a stop; always square off on the last trading day of the contract.
Metric	Value
total_trades	112
wins	40
losses	72
win_rate_pct	35.71428571428572
gross_profit	186950.8025588332
gross_loss	-101957.7231104605
net_pnl	84993.0794483727
net_points	3943.664817301098
avg_pnl_per_trade	758.8667807890419
max_win	33644.37910008551
max_loss	-6479.307056729425
max_drawdown	-44957.74063675845
reversals	88
partial_exits	3
stop_loss_exits	107

રિપોર્ટમાં **Summary** (કુલ આંકડા), **Trades** (દરેક ટ્રેડ), **Equity** (ઇક્વિટી કર્વ) અને **Events** (સ્ટ્રેટેજી ઘટનાઓ) — ચાર શીટ હોય છે.

Report — Trades sheet (first rows)

Trade ID	Cycle	Side	Lots	Entry Time	Entry Price	Exit Time	Exit Price	P&L Points	P&L (Rs)	Stop Level	Entry Reason	Exit Reason	Partial	Reversal	Status	Strategy
trade_6de510a5e8b9	cyc_1	BUY	1	2025-11-03 11:19:00	51,112.37	2025-11-03 11:19:00	51,111.37	-1	-348.31	51,088.14	entry_signal	stop_loss	False	False	CLOSED	v1_baseline
trade_f495e03cfc7f	cyc_1	BUY	1	2025-11-03 11:19:00	51,112.37	2025-11-03 11:19:00	51,111.37	-1	-348.31	51,088.14	entry_signal	stop_loss	False	False	CLOSED	v1_baseline
trade_adcafd5ba2c	cyc_1	SELL	1	2025-11-03 11:19:00	51,111.37	2025-11-03 12:43:00	51,265.51	-154.14	-4,942.85	51,211.21	reversal	stop_loss	False	True	CLOSED	v1_baseline
trade_aa6f6748b27f	cyc_1	SELL	1	2025-11-03 11:19:00	51,111.37	2025-11-03 12:43:00	51,265.51	-154.14	-4,942.85	51,211.21	reversal	stop_loss	False	True	CLOSED	v1_baseline

Trade ID	Cycle	Side	Lots	Entry Time	Entry Price	Exit Time	Exit Price	P&L Points	P&L (Rs)	Stop Level	Entry Reason	Exit Reason	Partial	Reversal	Status	Strategy
trade_62f633012aa9	cyc__1	BUY	1	2025-11-03 12:43:00	51,265.51	2025-11-03 13:14:00	51,279.49	13.98	100.26	51,211.21	reversal	stop_loss	False	True	CLOSED	v1_baseline
trade_1d6b6b8b76a0	cyc__1	BUY	1	2025-11-03 12:43:00	51,265.51	2025-11-03 13:14:00	51,279.49	13.98	100.26	51,211.21	reversal	stop_loss	False	True	CLOSED	v1_baseline
trade_e893a6bf8ad4	cyc__1	SELL	1	2025-11-03 13:14:00	51,279.49	2025-11-03 13:15:00	51,268.95	10.54	-3.00	51,211.21	reversal	stop_loss	False	True	CLOSED	v1_baseline
trade_5524932603f7	cyc__1	SELL	1	2025-11-03 13:14:00	51,279.49	2025-11-03 13:15:00	51,268.95	10.54	-3.00	51,211.21	reversal	stop_loss	False	True	CLOSED	v1_baseline
trade_bf4d7f6d39d8	cyc_2025-11-27_2	SELL	1	2025-12-03 14:18:00	48,583.41	2025-12-03 14:18:00	48,584.41	-1	-333.73	48,498.77	entry_signal	stop_loss	False	False	CLOSED	v1_baseline
trade_6923724c6bdc	cyc_2025-11-27_2	SELL	1	2025-12-03 14:18:00	48,583.41	2025-12-03 14:18:00	48,584.41	-1	-333.73	48,498.77	entry_signal	stop_loss	False	False	CLOSED	v1_baseline
trade_f73cb70740e1	cyc_2025-11-27_2	BUY	1	2025-12-03 14:18:00	48,584.41	2025-12-03 14:48:00	48,440.48	-143.93	-4,620.91	48,380.58	reversal	stop_loss	False	True	CLOSED	v1_baseline
trade_9baf49c65256	cyc_2025-11-27_2	BUY	1	2025-12-03 14:18:00	48,584.41	2025-12-03 14:48:00	48,440.48	-143.93	-4,620.91	48,380.58	reversal	stop_loss	False	True	CLOSED	v1_baseline

9. Step 6 — the web console (all 6 tabs)

A full control panel — not a read-only page. Start/stop sessions, edit strategies, run backtests, manage settings and hit the emergency stop.

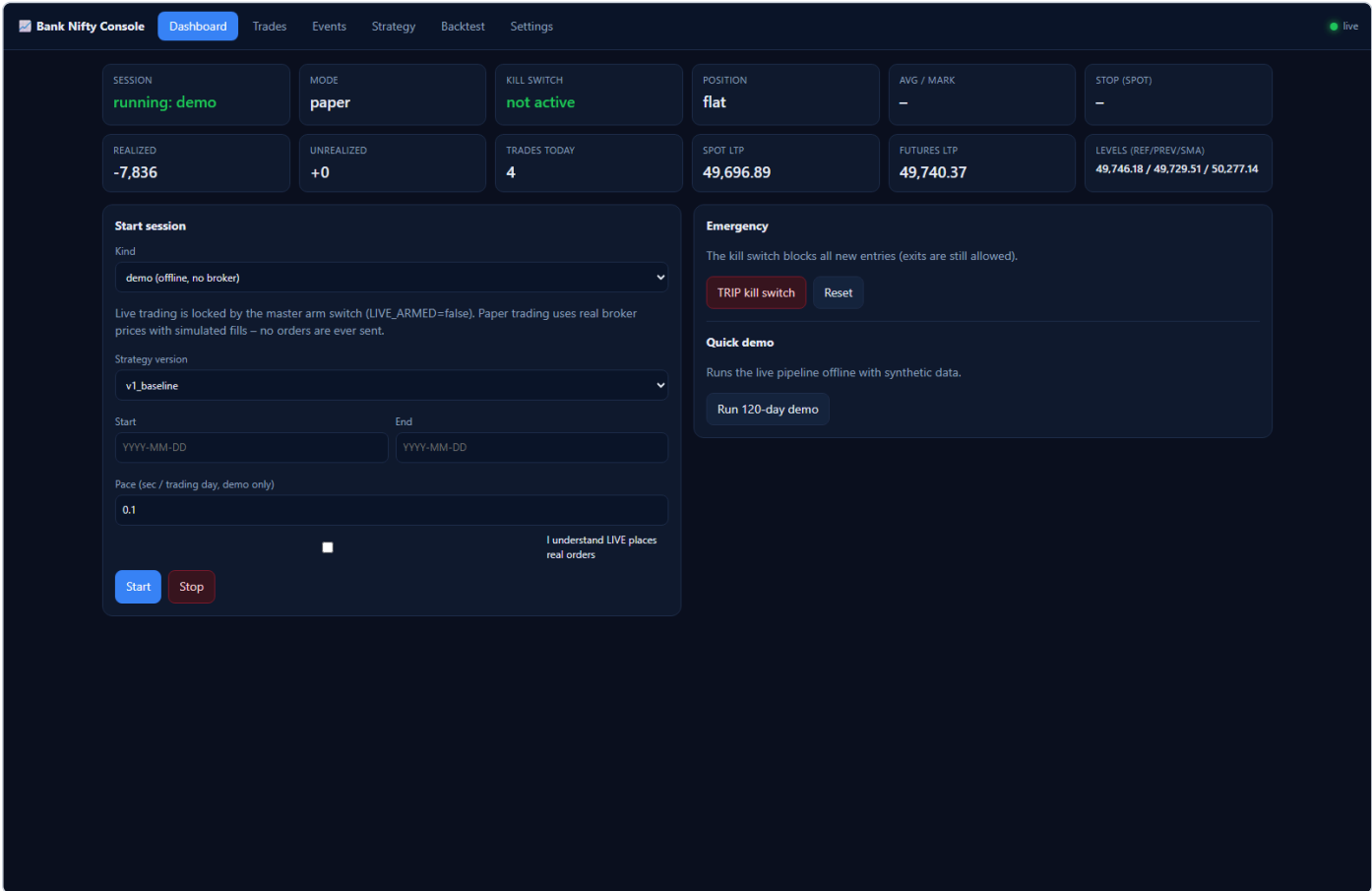
```
PowerShell: python -m banknifty_trading_app.main web

Bank Nifty web console: http://127.0.0.1:8080

No accounts yet - visit /register to create the first (admin) account
```

બ્રાઉઝરમાં `http://127.0.0.1:8080` ખોલો. ડાબી બાજુના ટેબ વડે Dashboard, Trades, Events, Strategy, Backtest અને Settings જોઈ શકાય. **Live** વિકલ્પ ત્યાં સુધી બંધ રહે છે જ્યાં સુધી CLI થી arm ન કરો.

Tab 1 — Dashboard



Dashboard: session, mode, kill switch, position, P&L, stop, Spot/Futures LTP, levels — updated live over WebSocket.

Start a session from the **Start session** panel: choose *demo* (offline), *paper* (real prices, simulated fills) or *LIVE* (locked). The checkbox and the code-level arm switch are two independent gates for live.

Tab 2 — Trades

Bank Nifty Console

Dashboard

Trades

Events

Strategy

Backtest

Settings

live

Refresh

100 rows

side	lots	entry	entry_px	exit	exit_px	pts	p&l	stop	exit_reason	rev	part	strategy
SELL	1	2026-10-04 10:05:59.962367	51,279.5	2026-10-04 10:05:59.962367	51,268.95	10.55	+316.5	51,211.21	stop_loss	yes		reversal
SELL	1	2026-10-04 10:05:59.962367	51,279.5	2026-10-04 10:05:59.962367	51,268.95	10.55	+316.5	51,211.21	stop_loss	yes		reversal
BUY	1	2026-10-04 10:05:59.962367	51,265.5	2026-10-04 10:05:59.962367	51,279.5	14	+420	51,211.21	stop_loss	yes		reversal
BUY	1	2026-10-04 10:05:59.962367	51,265.5	2026-10-04 10:05:59.962367	51,279.5	14	+420	51,211.21	stop_loss	yes		reversal
SELL	1	2026-10-04 10:05:59.956465	51,111.35	2026-10-04 10:05:59.962367	51,265.5	-154.15	-4,624.5	51,211.21	stop_loss	yes		reversal
SELL	1	2026-10-04 10:05:59.956465	51,111.35	2026-10-04 10:05:59.962367	51,265.5	-154.15	-4,624.5	51,211.21	stop_loss	yes		reversal
BUY	1	2026-10-04 10:05:59.956465	51,112.35	2026-10-04 10:05:59.956465	51,111.35	-1	-30	51,088.14	stop_loss			entry_signal
BUY	1	2026-10-04 10:05:59.956465	51,112.35	2026-10-04 10:05:59.956465	51,111.35	-1	-30	51,088.14	stop_loss			entry_signal
BUY	1	2026-10-04 10:03:45.018851	46,244.35	2026-10-04 10:03:45.020874	46,222.55	-21.8	-654	46,162.8	stop_loss	yes		reversal
BUY	1	2026-10-04 10:03:45.018851	46,244.35	2026-10-04 10:03:45.020874	46,222.55	-21.8	-654	46,162.8	stop_loss	yes		reversal
SELL	1	2026-10-04 10:03:45.010496	46,120.15	2026-10-04 10:03:45.016832	46,244.35	-124.2	-3,726	46,162.8	stop_loss			entry_signal
SELL	1	2026-10-04 10:03:45.010496	46,120.15	2026-10-04 10:03:45.016832	46,244.35	-124.2	-3,726	46,162.8	stop_loss			entry_signal
BUY	1	2026-10-04 10:03:37.804478	47,357.15	2026-10-04 10:03:37.804478	47,369.1	11.95	+358.5	47,302.08	stop_loss	yes		reversal
BUY	1	2026-10-04 10:03:37.804478	47,357.15	2026-10-04 10:03:37.804478	47,369.1	11.95	+358.5	47,302.08	stop_loss	yes		reversal
SELL	1	2026-10-04 10:03:37.792542	47,328.85	2026-10-04 10:03:37.803477	47,357.15	-28.3	-849	47,302.08	stop_loss	yes		reversal
SELL	1	2026-10-04 10:03:37.792542	47,328.85	2026-10-04 10:03:37.803477	47,357.15	-28.3	-849	47,302.08	stop_loss	yes		reversal
BUY	1	2026-10-04 10:03:37.790545	47,346	2026-10-04 10:03:37.791533	47,328.85	-17.15	-514.5	47,302.08	stop_loss	yes		reversal
BUY	1	2026-10-04 10:03:37.790545	47,346	2026-10-04 10:03:37.791533	47,328.85	-17.15	-514.5	47,302.08	stop_loss	yes		reversal
SELL	1	2026-10-04 10:03:37.403911	47,324.1	2026-10-04 10:03:37.789535	47,346	-21.9	-657	47,892.68	stop_loss	yes		reversal
SELL	1	2026-10-04 10:03:37.403911	47,324.1	2026-10-04 10:03:37.789535	47,346	-21.9	-657	47,892.68	stop_loss	yes		reversal
BUY	1	2026-10-04 10:03:37.400909	47,325.1	2026-10-04 10:03:37.402912	47,324.1	-1	-30	47,306.07	stop_loss			entry_signal
BUY	1	2026-10-04 10:03:37.400909	47,325.1	2026-10-04 10:03:37.402912	47,324.1	-1	-30	47,306.07	stop_loss			entry_signal
SELL	1	2026-10-04 10:03:27.548202	48,050.85	2026-10-04 10:03:27.548202	48,007.75	43.1	+1,293	47,958.63	stop_loss	yes		reversal

Trades: every closed leg with entry/exit, points, P&L, stop, exit reason, reversal and partial flags.

Tab 3 — Events

Bank Nifty ConsoleDashboardTradesEventsStrategyBacktestSettings

live

Refresh

TIME	TYPE	DETAIL	STRATEGY
2026-02-27 10:20:00	enter	SELL 2 lot(s) reason=reversal spot=46160.99 stop=46162.8	v1_baseline
2026-02-27 10:20:00	stop_loss	BUY stop hit at spot=46160.99 stop=46162.801290317635	v1_baseline
2026-02-27 10:20:00	enter	SELL 2 lot(s) reason=reversal spot=46160.99 stop=46162.8	v1_baseline
2026-02-27 10:20:00	stop_loss	BUY stop hit at spot=46160.99 stop=46162.801290317635	v1_baseline
2026-02-27 10:19:00	enter	BUY 2 lot(s) reason=reversal spot=46165.11 stop=46162.8	v1_baseline
2026-02-27 10:19:00	stop_loss	SELL stop hit at spot=46165.11 stop=46162.801290317635	v1_baseline
2026-02-27 10:19:00	enter	BUY 2 lot(s) reason=reversal spot=46165.11 stop=46162.8	v1_baseline
2026-02-27 10:19:00	stop_loss	SELL stop hit at spot=46165.11 stop=46162.801290317635	v1_baseline
2026-02-27 09:35:00	enter	SELL 2 lot(s) reason=entry_signal spot=46053.79 stop=46162.8	v1_baseline
2026-02-27 09:35:00	enter	SELL 2 lot(s) reason=entry_signal spot=46053.79 stop=46162.8	v1_baseline
2026-02-27 09:15:00	day_start	2026-02-27 ref=46062.87 prev=46162.80 sma=46907.4 last_day=False	v1_baseline
2026-02-27 09:15:00	day_start	2026-02-27 ref=46062.87 prev=46162.80 sma=46907.4 last_day=False	v1_baseline
2026-02-26 09:15:00	day_start	2026-02-26 ref=46920.74 prev=46062.87 sma=46993.89 last_day=True	v1_baseline
2026-02-26 09:15:00	day_start	2026-02-26 ref=46920.74 prev=46062.87 sma=46993.89 last_day=True	v1_baseline
2026-02-25 09:15:00	day_start	2026-02-25 ref=46973.98 prev=46920.74 sma=47055.68 last_day=False	v1_baseline
2026-02-25 09:15:00	day_start	2026-02-25 ref=46973.98 prev=46920.74 sma=47055.68 last_day=False	v1_baseline
2026-02-24 09:15:00	day_start	2026-02-24 ref=47292.63 prev=46973.98 sma=47062.94 last_day=False	v1_baseline
2026-02-24 09:15:00	day_start	2026-02-24 ref=47292.63 prev=46973.98 sma=47062.94 last_day=False	v1_baseline
2026-02-23 09:15:00	day_start	2026-02-23 ref=47202.22 prev=47292.63 sma=47052.48 last_day=False	v1_baseline
2026-02-23 09:15:00	day_start	2026-02-23 ref=47202.22 prev=47292.63 sma=47052.48 last_day=False	v1_baseline
2026-02-20 09:15:00	day_start	2026-02-20 ref=47237.63 prev=47202.22 sma=47037.58 last_day=False	v1_baseline
2026-02-20 09:15:00	day_start	2026-02-20 ref=47237.63 prev=47202.22 sma=47037.58 last_day=False	v1_baseline
2026-02-19 09:15:00	day_start	2026-02-19 ref=46795.35 prev=47237.63 sma=46977.13 last_day=False	v1_baseline
2026-02-19 09:15:00	day_start	2026-02-19 ref=46795.35 prev=47237.63 sma=46977.13 last_day=False	v1_baseline

Events: the strategy's own decision log (day start, entry, stop hit, reversal, square-off).

Tab 4 — Strategy editor

Bank Nifty Console

Dashboard

Trades

Events

Strategy

Backtest

Settings

live

Versions

v1_baseline

active

v2_example

+ New version

Editor — v1_baseline

```
# =====
# Strategy Version 1 - baseline (the original strategy)
# =====
# Edit this file to change strategy behaviour. No Python changes required for
# the common cases (SMA period, percentages, lot counts, filters, stops...).
# =====

version: v1_baseline
description: >
  Original Bank Nifty strategy: 20 SMA filter on daily Spot closes; entry on a
  fresh cross of the day-before-yesterday Spot close; 2 lots; yesterday's Spot
  close as the ratcheting stop; 1.7% partial on one lot; 2-lot reversal on a
  stop; always square off on the last trading day of the contract.

instrument:
  symbol: BANKNIFTY
  exchange: NFO
  spot_symbol: "NIFTY BANK"
  spot_exchange: NSE
  lots: 2
  lot_size: 30
  product_type: NRML
  contract: near month      # near month | next month
```

Save

Set active

Saving validates the YAML against the rule registry before writing. Rules are data - no code changes needed.

Strategy: versioned YAML files. Saving validates against the rule registry before writing — rules are data, no code changes.

અહીં YAML સ્ટ્રેટેજી સીધી બ્રાઉઝરમાં એડિટ કરી શકાય. સેવ કરતી વખતે એપ નિયમો ચેક કરે છે. નવી વર્ઝન બનાવવા + **New version** દબાવો.

Tab 5 — Backtest

Bank Nifty Console

Dashboard

Trades

Events

Strategy

Backtest

Settings

live

Run backtest

Strategy

v1_baseline

Start

YYYY-MM-DD

End

YYYY-MM-DD

☒

synthetic data (no broker)

Run

strategy v1_baseline
trades 112
net P&L +84,993.08
report pdf_assets\capture\reports\backtest_v1_baseline_20261004.xlsx

Reports

backtest_v1_baseline_20261004.xlsx

34.5 KB

demo_report.xlsx

20.7 KB

live_trades_report.xlsx

17.7 KB

Backtest: run a synthetic or historical backtest in the browser and download the Excel reports from the list.

Tab 6 — Settings

Bank Nifty Console

Dashboard

Trades

Events

Strategy

Backtest

Settings

live

Settings

ACTIVE_STRATEGY

v1_baseline

BACKTEST_INTRABAR

conservative

DASHBOARD_ENABLED

true

EXPIRY_SQUAREOFF_TIME

15:15

LIVE_ARMED

false (locked)

Master arm switch. Edit with the CLI: `main arm-live / main disarm-live`.

LIVE_TRADING

false

MARKET_CLOSE

15:30

MARKET_OPEN

09:15

MARKET_PROTECT_BUFFER_POINTS

0.5

MAX_DAILY_LOSS

MAX_LOTS

10

MAX_TRADES_PER_DAY

MODE

paper

ORDER_TYPE

MARKET

PAPER_SLIPPAGE_POINTS

0.5

PRODUCT_TYPE

NRML

STALE_TICK_TIMEOUT_S

5

WEB_HOST

127.0.0.1

WEB_PORT

8080

live_armed

false

Save settings

Angel One credentials

Stored in .env (git-ignored). Leave a field blank to keep the current value.

API key

Client code

A12345

PIN

TOTP secret

Web token (optional)

blank = no auth on localhost

Save credentials

Auth token (this browser)

only if WEB_TOKEN is set

Use token

Settings: safe, non-secret settings + the Angel One credential form. LIVE_ARMED is shown but cannot be changed here.

The console can **display** the arm state but **cannot change it** — `LIVE_ARMED` is deliberately excluded from the editable settings. Arming is CLI-only.

Emergency — kill switch

Bank Nifty Console

Dashboard

Trades

Events

Strategy

Backtest

Settings

live

SESSION

running: demo

MODE

paper

KILL SWITCH

ACTIVE

POSITION

SELL 2

AVG / MARK

47,369.1 / 47,369.1

STOP (SPOT)

—

REALIZED

-22,263

UNREALIZED

+ -0

TRADES TODAY

18

SPOT LTP

47,202.22

FUTURES LTP

47,292.86

LEVELS (REF/PREV/SMA)

46,795.35 / 47,237.63 / 46,977.13

Start session

Kind

demo (offline, no broker)

Live trading is locked by the master arm switch (LIVE_ARMED=false). Paper trading uses real broker prices with simulated fills – no orders are ever sent.

Strategy version

v1_baseline

Start

End

YYYY-MM-DD

YYYY-MM-DD

Pace (sec / trading day, demo only)

0.1

Start

Stop

I understand LIVE places real orders

Emergency

The kill switch blocks all new entries (exits are still allowed).

TRIP kill switch

Reset

Quick demo

Runs the live pipeline offline with synthetic data.

Run 120-day demo

Kill switch ACTIVE: blocks all new entries (exits are still allowed) across every session. Reset with 'Reset' or 'unkill'.

10. Step 7 — `run` paper trading

The real thing, minus the real order. It logs in to Angel One, streams real WebSocket ticks for Spot and futures, computes real daily closes and levels, and runs the real strategy/risk/order-manager pipeline — then simulates the fill at the last tick $\pm$ `PAPER_SLIPPAGE_POINTS`.

```
PowerShell
```

```
python -m banknifty_trading_app.main run # MODE=paper from .env
```

What happens on one tick

1. **Tick arrives** on the WebSocket thread → handed to the asyncio loop.
2. **Strategy engine** updates indicators and checks every rule (SMA filter, reference cross, stop, partial, trailing, expiry).
3. If a rule fires, it emits a **signal intent** (BUY/SELL, lots, reason).
4. **Risk manager** checks the kill switch, max lots, and daily loss/trade limits.
5. **Order manager** sizes the order, dedupes by intent id, and routes it.
6. The **executor** fills it: paper = simulated at last tick; live = REST call to Angel One.
7. The **position book & ledger** update; the **batched writer** persists orders, fills, trades and events to SQLite.
8. The **dashboard / web console** reflects the new state over WebSocket.

દરેક tick પર: ભાવ આવે → સ્ટ્રેટેજી નિયમો ચેક થાય → સિગ્નલ બને → રિસ્ક ચેક થાય → ઓર્ડર બને → paper માં નકલી ફિલ થાય (live માં સાચો ઓર્ડર જાય) → P&L અપડેટ → ડેટાબેઝમાં સેવ થાય.

Paper trade for several full sessions, including an expiry day, before considering live. Watch the console, `logs/`, and the Excel report. Paper fills are optimistic — real slippage is usually worse.

11. Step 8 — go-live safety

Live trading requires **all three** flags, enforced in three independent places. If any one is missing, no real order can go out.

Flag	Meaning	Default
MODE=live	the session is a live session	paper
LIVE_TRADING=true	the operator opted in	false
LIVE_ARMED=true	master arm switch (CLI only)	false

Check the state any time

```
PowerShell: python -m banknifty_trading_app.main live-status

Live-trading interlock
=====
[-- ] MODE           paper
[-- ] LIVE_TRADING   false
[-- ] LIVE_ARMED     false

State: PAPER (safe)
live trading is LOCKED: LIVE_ARMED=false (master arm switch), MODE='paper' (need MODE=live), LIVE_TRADING=false (need LIVE_TRADING=true)
```

Preflight — the go-live checklist (read-only)

```
PowerShell: python -m banknifty_trading_app.main preflight

=====
GO-LIVE PREFLIGHT - read-only, places no orders
strategy=v1_baseline mode=paper armed=False
=====

[1. Live-trading interlock]
INFO MODE=paper
INFO LIVE_TRADING=False
INFO LIVE_ARMED=False
PASS Live trading is locked - no real orders can be placed. Good while paper trading.

[2b. Network / TLS]
PASS verifying TLS against the OS certificate store
INFO TLS is re-signed by 'Zscaler Inc.' (network inspection). Verification passes via the OS store; the proxy can see this traffic like any other HTTPS.
PASS instrument master reachable (33023 KB)

[2. Angel One account]
FAIL credentials missing: angel_api_key, angel_client_code, angel_pin, angel_totp_secret (run: setup)
INFO SmartAPI app must be ACTIVE in the Angel One portal (not deleted/expired)
INFO Your static IP must be whitelisted if the API key is IP-locked
INFO TOTP secret must be the base32 seed, not a rotating 6-digit code
INFO Contract notes / MIS enabled in the account if you use intraday product
PASS requests will report X-ClientPublicIP=167.103.138.83 (local 192.168.29.187)
WARN the SDK has not been patched yet (it still holds its built-in IP). It is patched automatically on login - set CLIENT_PUBLIC_IP to be sure.

[3. Risk limits]
PASS MAX_LOTS=10 allows the strategy's 2 lots
WARN MAX_DAILY_LOSS is not set: no automatic stop for a losing day
WARN MAX_TRADES_PER_DAY is not set: no cap on how many times it may re-enter
PASS kill switch is not tripped
INFO Verify the kill switch works: run 'kill test', confirm trading halts, 'unkill'

[4. Instrument & contract]
PASS trading BANKNIFTY on NFO, signals from NIFTY BANK (NSE)
PASS 2 lots x lot_size 30 = 60 qty per entry
WARN lot_size=30 is set in YAML - CONFIRM the current NSE BANKNIFTY lot size. A wrong lot size silently sends the wrong quantity.
PASS reversal on: 2 lots, cap 3/day, cooldown 60.0s

[5. Session window (IST)]
PASS MARKET_OPEN=09:15
PASS MARKET_CLOSE=15:30
PASS expiry square-off at 15:15, before the 15:30 close
INFO PRODUCT_TYPE=NRML carries overnight - you need overnight margin

[6. Execution model]
PASS ORDER_TYPE=MARKET: fills at best available price, no price control
PASS paper fills simulate 0.5 pt slippage; live fills are whatever the exchange gives you - usually worse
PASS order status timeout 10.0s (no blind fill assumption)
WARN the '2% move' trailing rule is an undecided placeholder - decide its formula in the strategy YAML before trading it live

[7. Operations (only you can confirm)]
INFO Machine stays ON and connected for the whole 09:15-15:30 window
INFO No OS sleep / hibernate / screen-lock that pauses the process
INFO Reliable internet; the WebSocket auto-reconnects but gaps lose ticks
INFO Angel One must allow unattended API order placement for your account
INFO Broker-side RMS / risk limits are not tighter than this app's limits
INFO You know how to flatten the position manually from the broker terminal
```

```
INFO You have watched paper results for several sessions and trust the P&L
INFO Account funds cover margin for the worst case, including reversal chains

[8. Broker session]
INFO skipped - add --online to test login and instruments

=====
VERDICT: NOT READY - 1 blocking issue(s):
- credentials missing: angel_api_key, angel_client_code, angel_pin, angel_totp_secret (run: setup)

Fix these before running 'arm-live'.
=====
```

Preflight દરેક વસ્તુ ચેક કરે છે: credentials, TLS/નેટવર્ક, lot size, રિસ્ક લિમિટ, કિલ્ સ્વિચ, અને બ્રોકર પોઝિશન. જ્યાં સુધી કોઈ FAIL હોય ત્યાં સુધી `arm-live` ન કરો. દરેક વસ્તુ **PASS / WARN / FAIL / INFO** માં ગ્રેડ થાય છે.

Arm / disarm & the kill switch

```
PowerShell

# 1. Arm the interlock (asks you to type ARM to confirm)
python -m banknifty_trading_app.main arm-live

# 2. Put it fully live in .env, then start:
#   MODE=live
#   LIVE_TRADING=true
python -m banknifty_trading_app.main run --live

# Emergency: block all new entries (exits still allowed)
python -m banknifty_trading_app.main kill "manual stop"
python -m banknifty_trading_app.main unkill

# Lock everything again -> LIVE_ARMED=false, LIVE_TRADING=false, MODE=paper
python -m banknifty_trading_app.main disarm-live
```

Before you arm live: preflight clean, you know how to flatten manually from the broker terminal, `lot_size` in the YAML matches the exchange, you funded the account for the worst case (including the reversal chain), and the machine stays awake 09:15–15:30 IST.

12. Step 9 — end-to-end trade walkthrough

One concrete example of how a signal becomes a trade, using the baseline rules.

1. At market open the engine fetches the last daily Spot closes and computes the **reference close** (close from 2 days ago), **yesterday's close** and the **20-day SMA**.
2. Spot **crosses above** the reference close **and** Spot > SMA → **BUY 2 lots** of BANKNIFTY futures. (Below → SELL 2 lots.)
3. The initial stop is yesterday's Spot close, used only if it is genuinely *protective* of the entry.
4. From the next day the stop uses the previous day's close and only **ratchets** (never loosens).
5. A **+1.7%** favourable move squares off **one** lot, carries the other, and the stop moves to breakeven.
6. If the stop is hit, the position exits and **reverses 2 lots** the other way, under the same rules (with a cooldown and a daily cap).
7. On the **last trading day** of the contract it squares off and carries nothing overnight.
8. Every decision, order, fill and trade is written to SQLite and the trade log, then surfaced in the console and the Excel report.

ઉદાહરણ: Spot ભાવ reference ભાવ ઉપર જાય + SMA ઉપર હોય તો BUY 2 lots. સ્ટોપ ગયા તો exit અને ઊલટું 2 lots (reversal). +1.7% થાય તો એક lot book કરો, બીજું ચાલુ રાખો. છેલ્લા દિવસે બધું બંધ કરો.

The 15 rules as implemented

#	Rule	Implementation
1	Record recent daily Spot closes	fetches at start-up; used for levels + SMA
2	20 SMA filter	daily Spot closes fixed at prior close (no look-ahead)
3	BUY only above SMA / SELL only below	<code>SmaFilter</code>
4	Cross day-before-yesterday close → BUY 2 lots	<code>ReferenceCrossEntry</code> , offset 2
5	Cross below it → SELL 2 lots	same rule, opposite side
6	Only one initial trade per cycle	<code>initial_trade_done</code> guard
7	Trade day: yesterday's close = stop	<code>DailyCloseStop.initial_stop</code>
8	From next day: previous day's close, ratchet only	<code>daily_stop(ratchet=True)</code>
9	+1.7% → square off 1 lot, carry 1, trail	<code>PercentPartialTarget</code>
10	Stop hit → exit and reverse 2 lots	<code>_stop_out</code> + reversal entry
11	Continue same rules after reversal	reversal uses the same rule objects
12	"2% move" trailing rule	configurable placeholder (no invented formula)
13	Last trading day → square off, no carry	<code>AlwaysSquareOff</code>
14	Record everything	SQLite + trade log
15	One-year backtest → Excel	<code>backtest</code> + reports/excel

Known ambiguity: the "2% move" trailing rule is undefined in the notes. It ships as `percent_move_placeholder` with `mode: disabled`. `custom` refuses to load until you implement it. Decide the formula before trading it live.

13. Troubleshooting & FAQ

"CERTIFICATE_VERIFY_FAILED" on every HTTPS call

A corporate proxy (Zscaler, Netskope, Fortinet...) re-signs TLS with its own root CA. That CA is in the Windows store but not in `certifi`. The app installs `truststore` at start-up so verification uses the OS store — verification stays **ON**. Run `preflight` and look at the **Network / TLS** section: it names the issuer. Disable with `USE_SYSTEM_TRUST_STORE=false`.

Login fails because of IP whitelisting

`smartapi-python` hardcodes its `X-ClientPublicIP` header. The app patches it with your real public IP before login. If your outbound IP **changes between requests** (multi-WAN / corporate proxy), no whitelist can work — ask Angel One to remove the IP restriction, or run on a VPS with a stable address.

"sb.append is not a function" / a smartapi import error

`smartapi-python` does not declare all of its own dependencies. `requirements.txt` pins `logzero` and `six` explicitly because a missing one makes the SDK fail at import.

Do I need an Angel One account to try it?

No. `demo`, `backtest --synthetic`, `report` and the `web` console all run fully offline. Credentials are needed only for `check`, paper `run` and live.

How do I change the strategy?

Edit YAML in `strategy_versions/` — e.g. 20 → 30 SMA (`filters[0].period: 30`), 1.7% → 2% (`partial.percent: 2.0`), 2 → 3 lots (`instrument.lots: 3`). Add a new file and set `ACTIVE_STRATEGY=v2_myrules` in `.env`. Nothing else changes.

How do I know the code is safe?

Paper by default · three-flag live interlock enforced in three places · the web console cannot arm live · duplicate-order protection · order-status confirmation with timeout · WebSocket auto-reconnect · crash recovery + position reconciliation at start-up · stale market-data detection · emergency kill switch · maximum-lot protection · full logging.

યાદ રાખો: આ ફાઇનાન્શિયલ એડવાઇસ નથી. પહેલાં backtest અને paper trading કરો. Live arm કરતાં પહેલાં preflight સાફ હોવું જોઈએ.

Quick reference

Command	What it does	Needs broker?
validate	show resolved strategy + config	no
setup	write credentials to .env (no echo)	no
check	connectivity self-test (no orders)	yes
demo	offline synthetic run through the live pipeline	no
backtest --synthetic	one-year strategy replay → Excel	no
backtest --start .. --end ..	historical backtest → Excel	yes
report	regenerate Excel from SQLite	no
web	full read-write console at 127.0.0.1:8080	no
run	paper (default) / live per .env	yes
preflight [--online]	go-live checklist (read-only)	optional
arm-live / disarm-live	arm / lock the live interlock	no
live-status	show the interlock state	no
kill "why" / unkill	emergency stop / reset	no